Experiment:- 6

Problem Statement:-

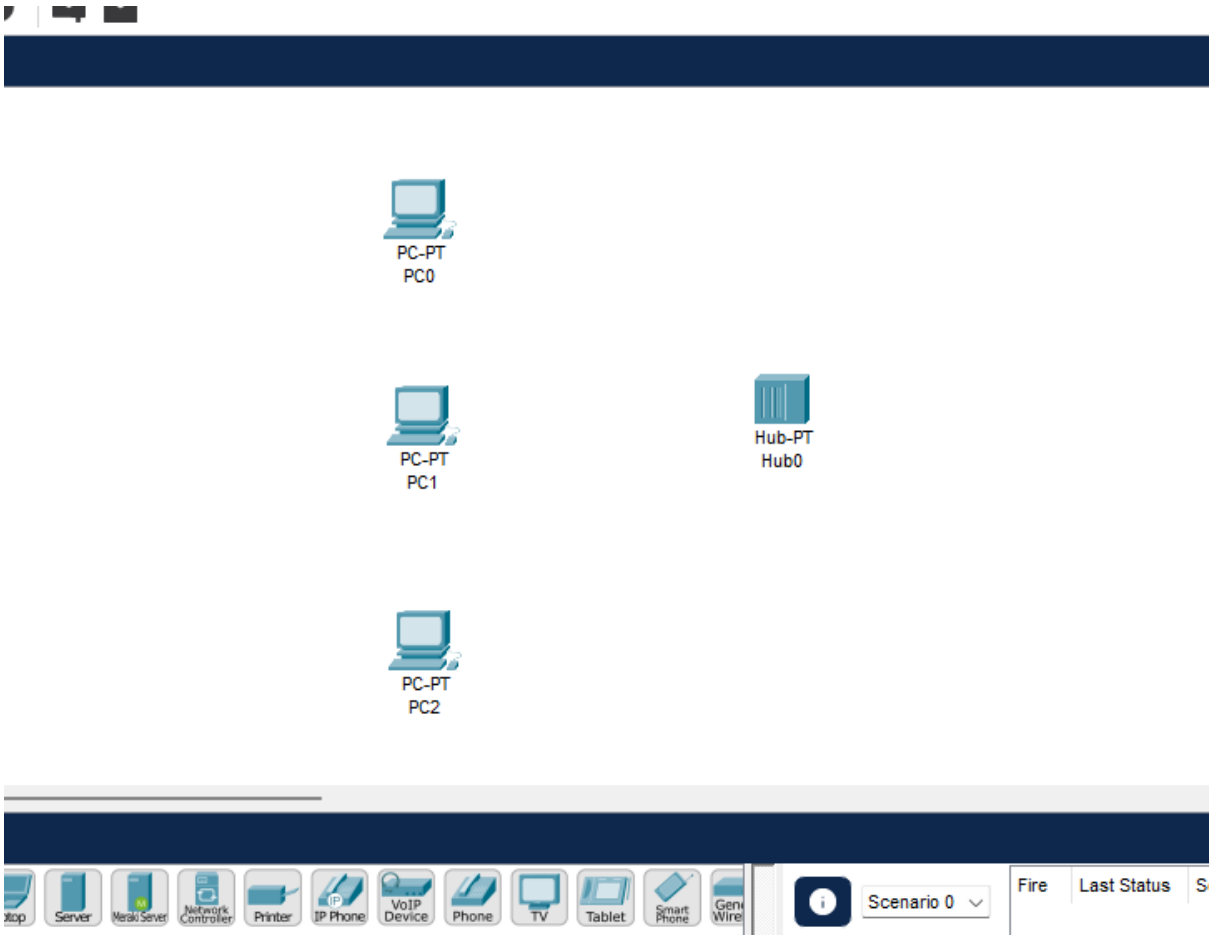
Develop a simulation to demonstrate multiple access protocols such as Pure ALOHA, Slotted ALOHA, CSMA/CD, and CSMA/CA using Cisco Packet Tracer. Analyze the performance of each protocol in handling network collisions and maximizing data transmission efficiency.

Aim:- Simulation of Multiple Access Protocols

PROCEDURE:- Pure Aloha

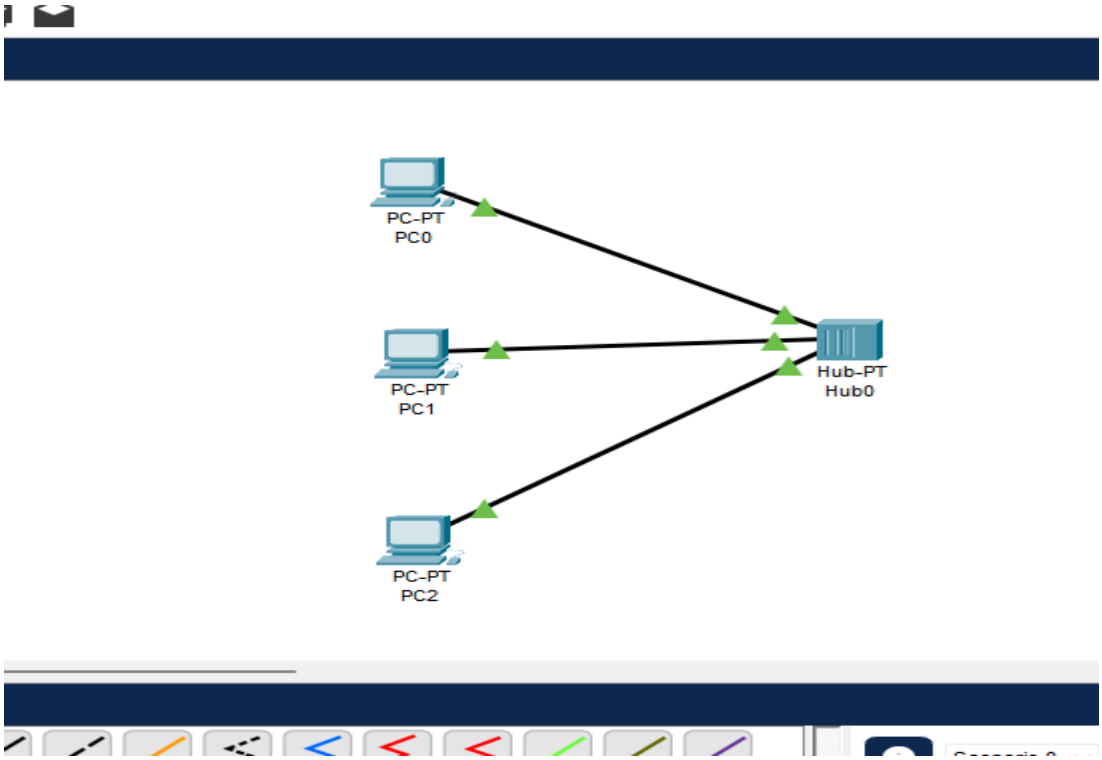
Step1:- Open Cisco Packet Tracer and create a new project.

Add devices:



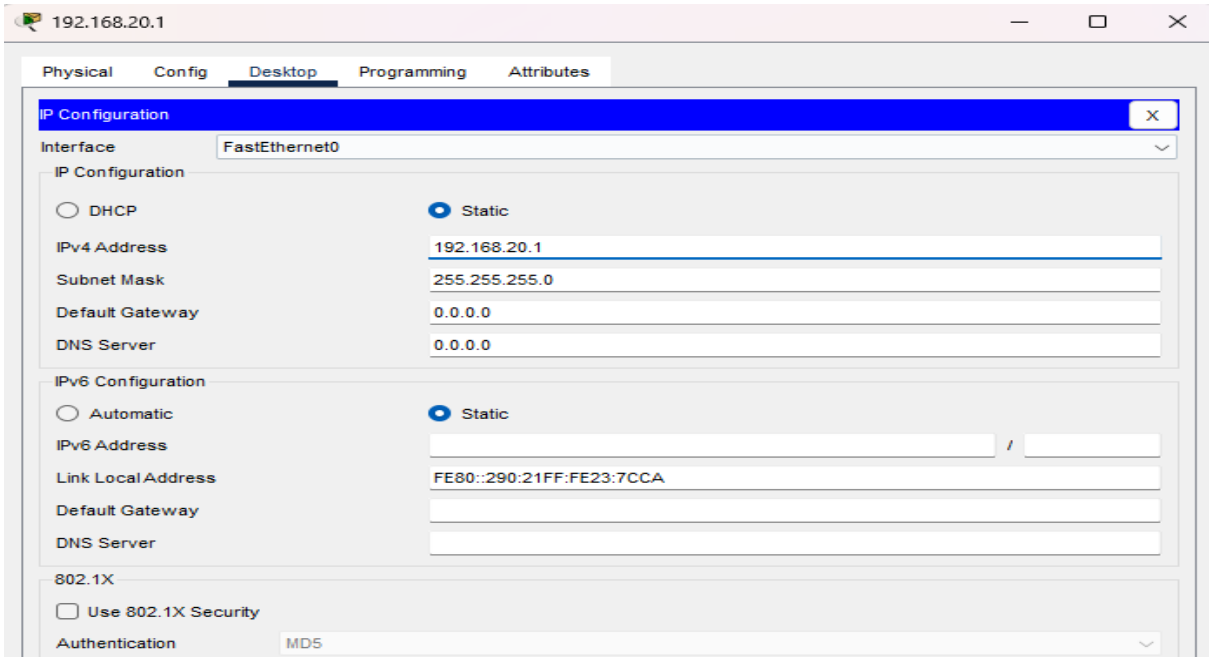
Step 2:- Connect devices:

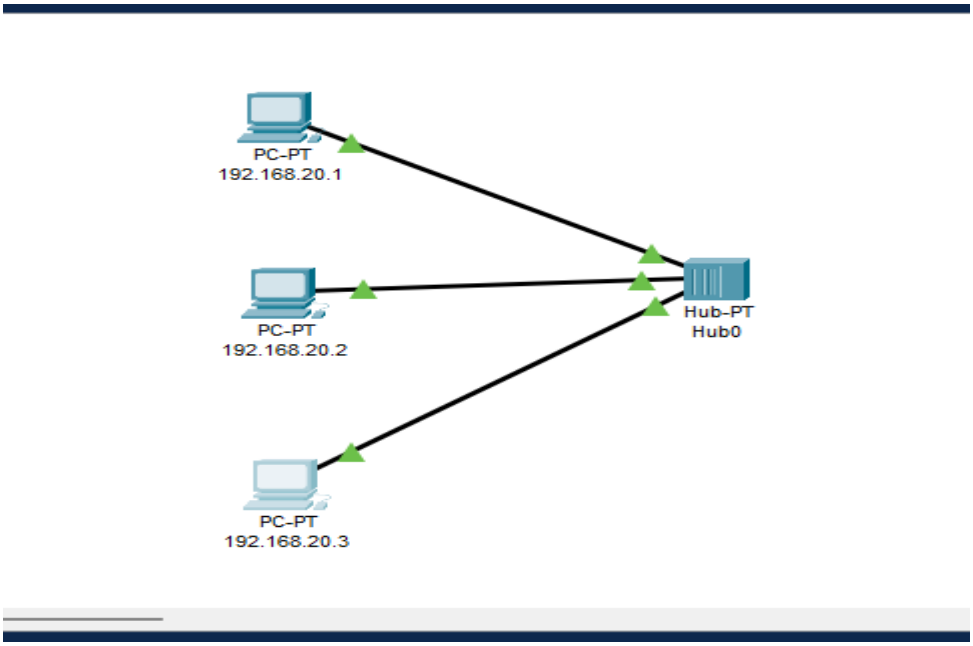
- Use **Copper Straight-Through Cable**
- Connect all PCs to the hub



Step 3:- Configure IP addresses:

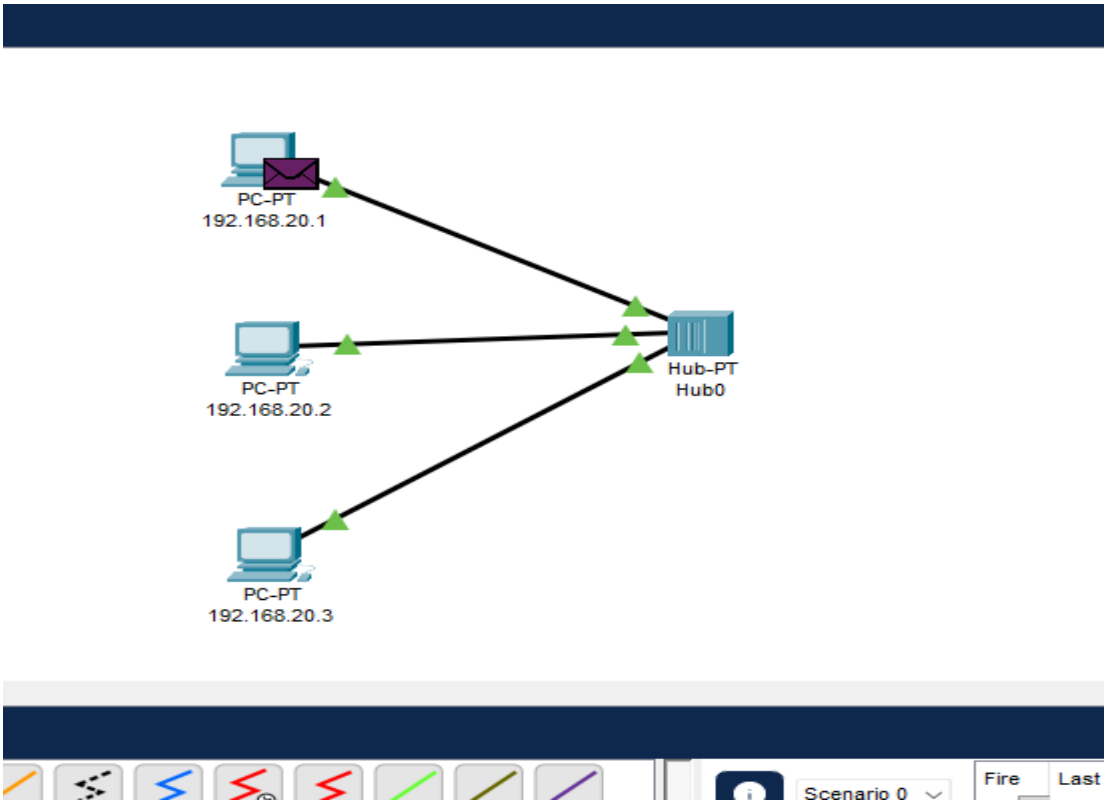
- Assign IP addresses to each PC in the same network





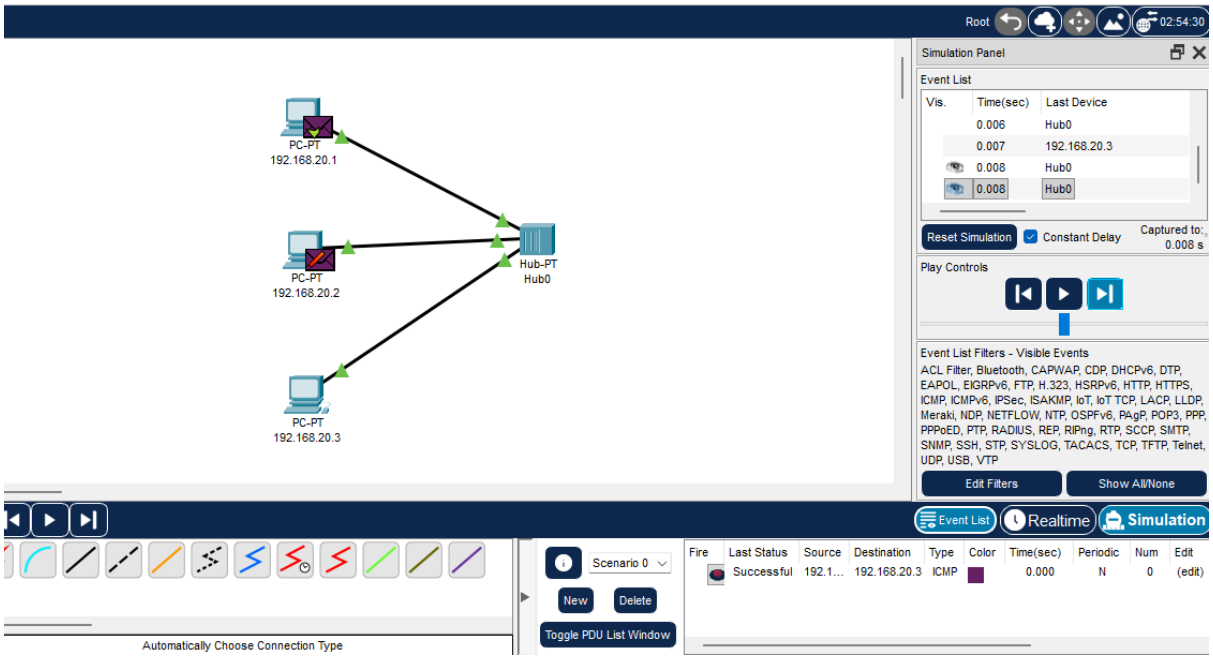
Step 4:- Simulate data transmission:

- Use **Add Simple PDU tool (envelope icon)**
- Send packets between different PCs



Step 5:-Enable Simulation Mode:

- Switch to **Simulation Mode**
- Click **Auto Capture/Play**



Step 6:- Observation:

Packet Flow through Hub:
In, the hub broadcasts packets to all devices, but only the destination accepts it.

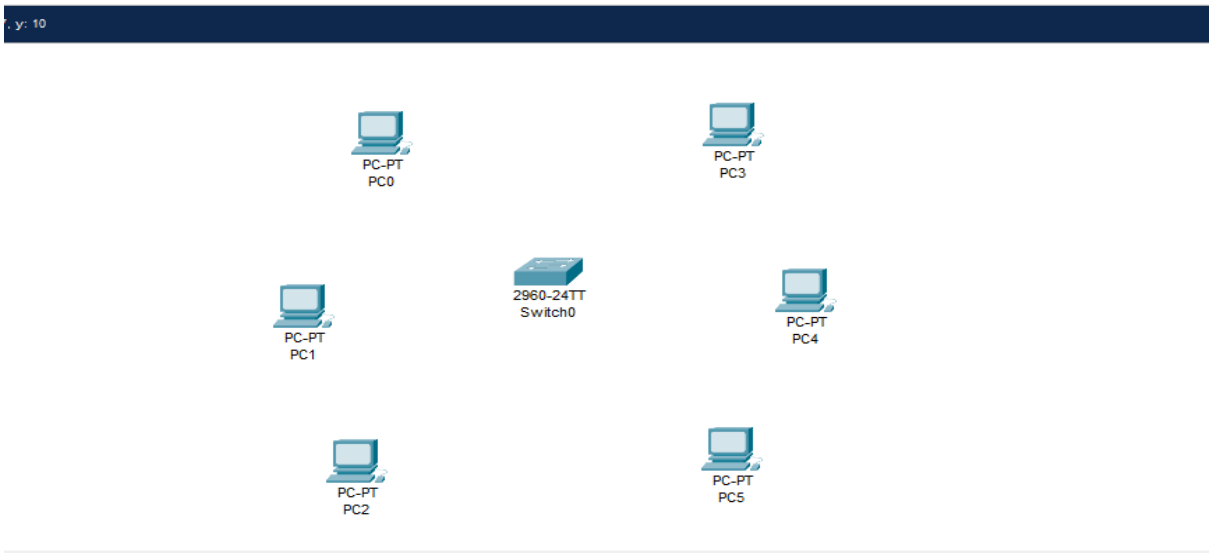
Collisions between Packets:
When multiple devices send data simultaneously, packets overlap, causing collisions.

Retransmissions after Collision:
After collision, data is lost and devices retransmit after a random delay.

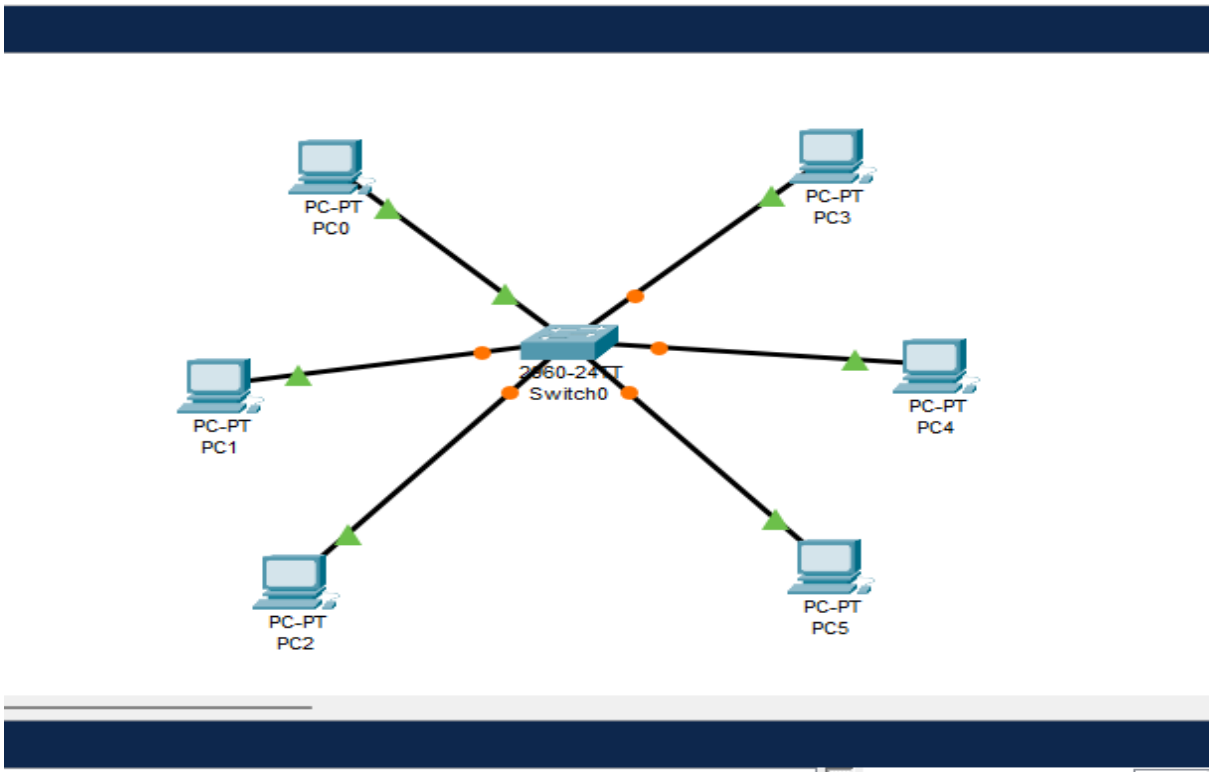
CSMA/CD (Switch-Based Network)

Procedure

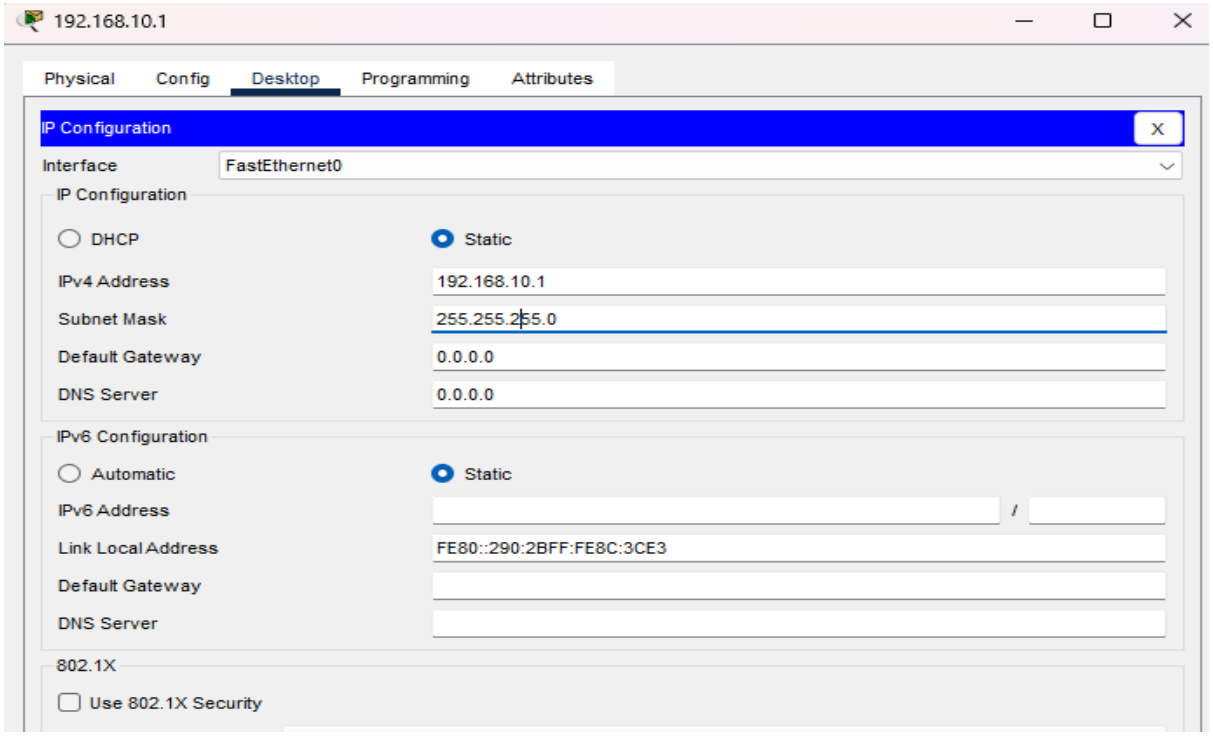
1. Open Cisco Packet Tracer and create a new project.
2. Add devices:
    - Drag and drop **4–6 PCs**
    - Add **1 Switch (2960)**

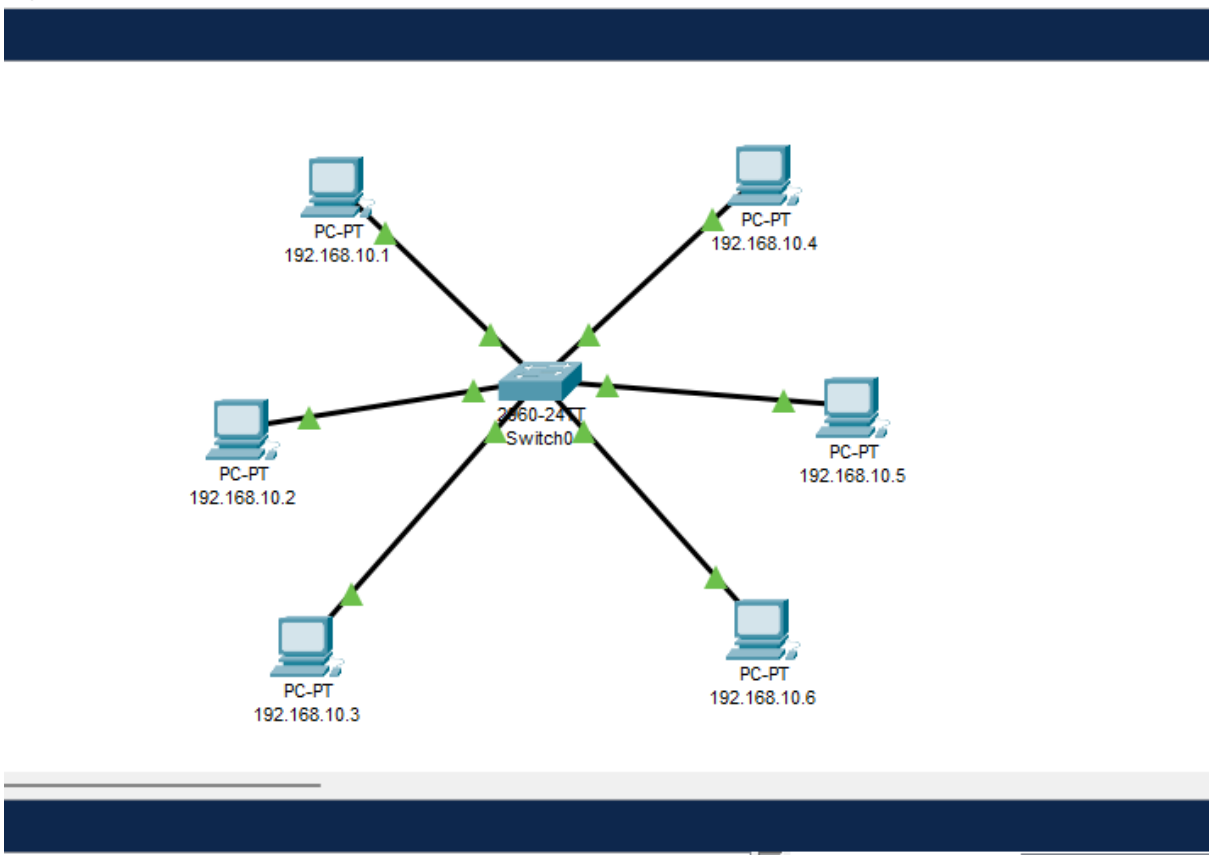


3. Connect devices:
    - Use **Copper Straight-Through Cable**
    - Connect all PCs to the switch

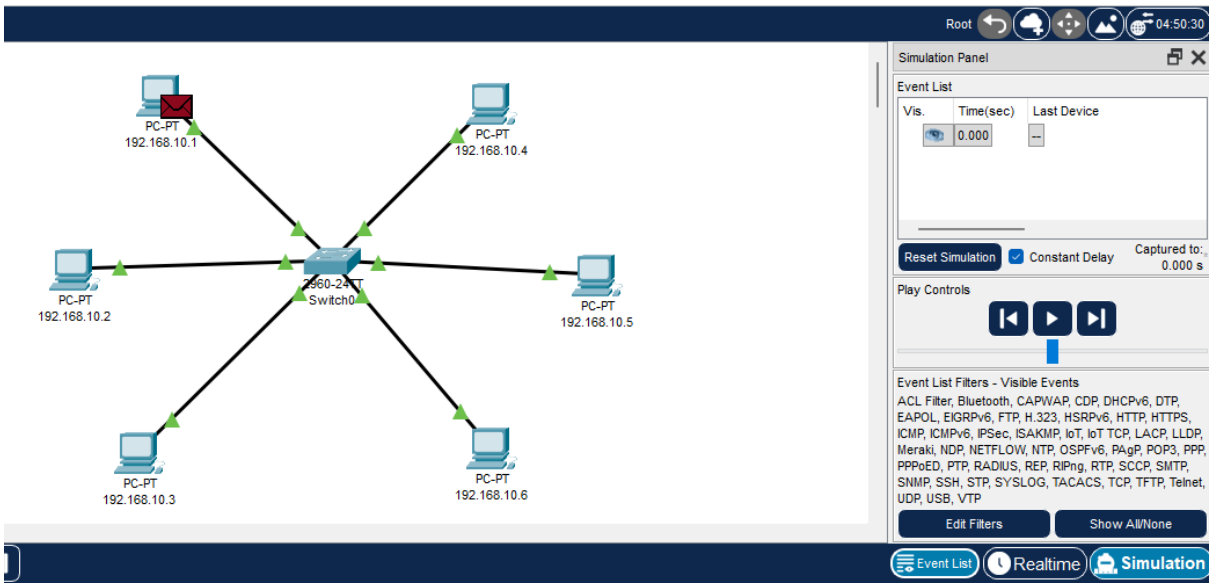


4. Configure IP addresses:
    - Assign IP addresses in the same network





5. Simulate data transmission:
    - Use **Add Simple PDU tool**
    - Send packets between different PCs



6. Observation:-

- **Packet flow through switch:** Data frames are forwarded through the switch to the correct destination port using MAC addresses.
- **Collision handling:** In **CSMA/CD**, devices detect collisions and retransmit after a random backoff time.

- **Data transmission efficiency:** Efficiency improves as collisions are minimized, but decreases under heavy network traffic.